Fall 2013 McNabb GDCTM Contest Answer Sheet
Middle School

Name: *KEY* School: _____ Date: _____

Contest Taking (Circle One):

(Pre-algebra) Algebra I

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 45 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. 8	6. 9	11. 9
2. 2	7. -1	12. 11
3. 144	8. 4	13. 4
4. 40,000	9. 8	14. 5^{18}
5. 9	10. Saturday	15. 17th

SCORE:_____

Fall 2013 McNabb GDCTM Contest Answer Sheet
Middle School

Name: *KEY* School: _____ Date: _____

Contest Taking (Circle One):

Pre-algebra (Algebra I)

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 45 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. 2	6. $12p+7q$	11. 10
2. -1	7. 2	12. 18
3. 11	8. $\frac{1}{6}+\frac{1}{102}$ or others	13. $3a-5b$
4. 19	9. 21	14. No
5. 100(%)	10. $\frac{128y^2}{9x}$	15. 19,683

SCORE:_____

Fall 2013 McNabb GDCTM Contest Answer Sheet
Upper School Contests

Name: * KEY * School: _____ Date: _____

Contest Taking (Circle One):

(Geometry) Algebra II Pre-calculus Calculus

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 60 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. 100 (%)	6. 289	11. 286
2. 2	7. 3:1 or 3	12. $2\sqrt{37}$
3. $\frac{5a+5b}{10}$ or $\frac{a+b}{2}$	8. 3	13. 71
4. 360	9. 50	14. (23,9) or any of form (23+31k, 9+50k)
5. $32\frac{1}{3}$	10. -7	15. $\frac{2}{3}, \frac{3}{2}$

SCORE: _____

Fall 2013 McNabb GDCTM Contest Answer Sheet
Upper School Contests

Name: *KEY* School: _____ Date: _____

Contest Taking (Circle One):

Geometry (Algebra II) Pre-calculus Calculus

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 60 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. $\frac{128y^2}{9x}$	6. 3:1 or 3	11. 6
2. NO	7. $2/3, 3/2$	12. 2
3. 21	8. 71	13. 575
4. $32/3$	9. $a \geq 1/2$ or $[1/2, \infty)$	14. III
5. 3	10. 310	15. (35, 4)

SCORE: _____

Fall 2013 McNabb GDCTM Contest Answer Sheet
Upper School Contests

Name: *KEY* School: _____ Date: _____

Contest Taking (Circle One):

Geometry Algebra II (Pre-calculus) Calculus

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 60 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. 9	6. 3	11. 0
2. NO	7. $a \geq 1/2$ or $[1/2, \infty)$	12. 33
3. 10	8. (23, 9) or any of form $(23+31k, 9+50k)$	13. 6
4. 18	9. 31	14. 19,683
5. 286	10. 2304	15. 3

SCORE:_____

Fall 2013 McNabb GDCTM Contest Answer Sheet
Upper School Contests

Name: *KEY* School: _____ Date: _____

Contest Taking (Circle One):

Geometry Algebra II Pre-calculus (Calculus)

DIRECTIONS:

1. You are NOT permitted to use a calculator or any other electronic device. You have 60 minutes.
2. All answers should be in standard simplified from unless otherwise stated. All variables are assumed to stand for real numbers unless otherwise stated.

1. 11	6. III	11. $4 \cdot \sqrt[4]{27}$
2. 10	7. 7	12. 48
3. 289	8. 2, -7	13. $f(x) = x^2 - 4x + 2$
4. 2304	9. $(-4/3, 4)$	14. 6
5. 286	10. 16	15. -2

SCORE: _____